

lab 10

subnetting

Manual Subnetting Implementation 192.192.192.0/29

Parameter	Value
Subnet Mask	255.255.255.248
Binary Mask	11111111.11111111.11111111.1111000
Network Bits	29
Host Bits	3
Hosts per Subnet	$2^3 - 2 = 6$ usable hosts
Total Subnets	$2^{21} = 2,097,152$
Block Size	8

Subnet #	Network Address	Usable Range	Broadcast	Purpose
1	192.192.192.0/29	192.192.192.1 - 192.192.192.6	192.192.192.7	LAN 1 (Router1 + 4 PCs)
2	192.192.192.8/29	192.192.192.9 - 192.192.192.14	192.192.192.15	LAN 2 (Router2 + 4 PCs)
3	192.192.192.16/29	192.192.192.17 - 192.192.192.22	192.192.192.23	LAN 3 (Router0 + 4 PCs)
4	192.192.192.24/29	192.192.192.25 - 192.192.192.30	192.192.192.31	WAN Link R0-R1
5	192.192.192.32/29	192.192.192.33 - 192.192.192.38	192.192.192.39	WAN Link R0-R2

```

Router(config)#int fa0/0
Router(config-if)#ip address 192.192.192.1 255.255.255.248
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface se0/0
%Invalid interface type and number
Router(config)#interface se2/0
Router(config-if)#ip address 192.192.192.9 255.255.255.248
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#
Router(config-if)#exi
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.192.192.0
Router(config-router)#exit

```

```

Router1
Physical Config CLI Attributes
IOS Command Line Interface

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.192.192.17 255.255.255.248
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip address 192.192.192.10 255.255.255.248
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config)#router rui
^
% Invalid input detected at '^' marker.

Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.192.192.0
Router(config-router)#exit

```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 191.191.16.1 255.255.248.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#ex
Router(config)#int se2/0
Router(config-if)#ip address 191.191.24.6 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
```

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 191.191.8.1 255.255.248.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#ex
Router(config)#int se2/0
Router(config-if)#ip address 191.191.24.2 255.255.255.252
Router(config-if)# no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#e
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

% Ambiguous command: "e"
Router(config-if)#ex
Router(config)#int se3/0
Router(config-if)#ip address 191.191.24.5 255.255.255.252
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
```

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 1
Router(config-router)#neighbor 191.191.24.2 remote-as 2
Router(config-router)#network 191.191.0.0 mask 255.255.248.0
Router(config-router)#network 191.191.254.1 mask 255.255.255.255
Router(config-router)#ex
```

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.192.192.18

Pinging 192.192.192.18 with 32 bytes of data:

Reply from 192.192.192.18: bytes=32 time=10ms TTL=126
Reply from 192.192.192.18: bytes=32 time=1ms TTL=126
Reply from 192.192.192.18: bytes=32 time=1ms TTL=126
Reply from 192.192.192.18: bytes=32 time=8ms TTL=126

Ping statistics for 192.192.192.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 5ms
```

part 2 subnetting

Company VLSM Subnetting 200.16.100.0/24

Sort requirements by size (descending)

1. Development: 32 PCs
2. Finance: 27 PCs

3. HR: 10 PCs

Calculate required host bits for each

Formula: $2^h - 2 \geq \text{required hosts}$

Department	Required Hosts	Host Bits (h)	Usable Hosts	Subnet Mask	CIDR
Development	32	6 ($2^6 - 2 = 62$)	62	255.255.255.192	/26
Finance	27	5 ($2^5 - 2 = 30$)	30	255.255.255.224	/27
HR	10	4 ($2^4 - 2 = 14$)	14	255.255.255.240	/28

Department	Network Address	Usable Range	Broadcast	Default Gateway
Development	200.16.100.0/26	200.16.100.1 - 200.16.100.62	200.16.100.63	200.16.100.1
Finance	200.16.100.64/27	200.16.100.65 - 200.16.100.94	200.16.100.95	200.16.100.65
HR	200.16.100.96/28	200.16.100.97 - 200.16.100.110	200.16.100.111	200.16.100.97

part 3

```

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 195.168.10.33 255.255.255.224
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#int se2/0
Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip address 195.168.10.98 255.255.255.252
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
exit
Router(config)#int se3/0
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#ip address 195.168.10.101 255.255.255.252
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 195.168.10.0

```

IOS Command Line Interface

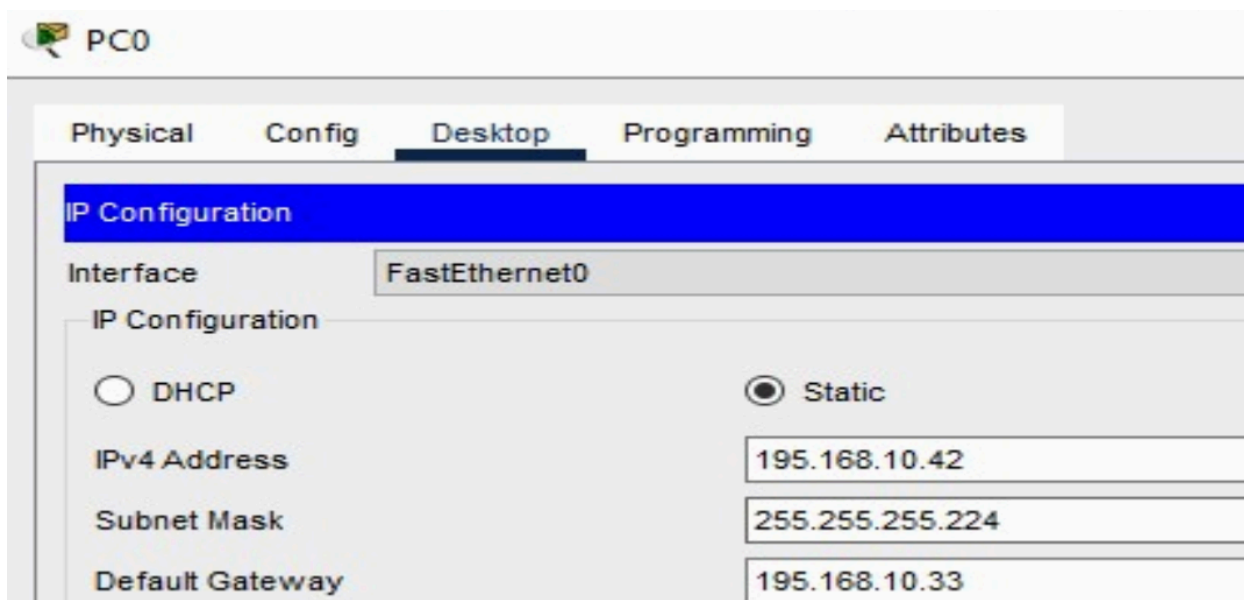
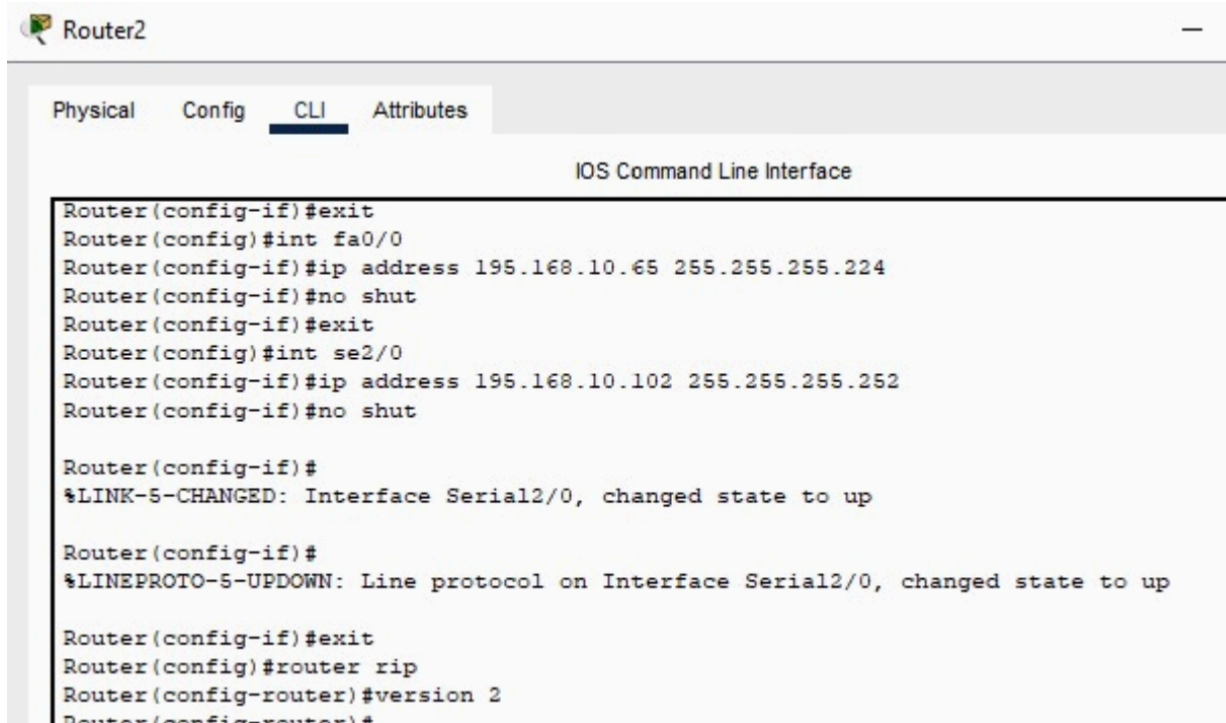
```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 195.168.10.1 255.255.255.224
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int se2/0
Router(config-if)#ip address 195.168.10.97 255.255.255.252
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 195.168.10.0
```

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter ar
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    195.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
R       195.168.10.0/27 [120/1] via 195.168.10.97, 00:00:04, Serial2/0
C       195.168.10.32/27 is directly connected, FastEthernet0/0
C       195.168.10.96/30 is directly connected, Serial2/0
C       195.168.10.100/30 is directly connected, Serial3/0
```

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface **FastEthernet0**

IP Configuration

☐ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway